

Dive into the Abyss: Submarine Squad on Your Guard!

Summary

MCM has launched the "Submersible Journey - The Ionian Wreck Search" campaign. As MCM security engineers, we provide a search model of the missing submersible to provide safety protection for the passengers.

For question 1, this paper establishes a dynamic prediction model for the position of the submersible. Step1: Analyze the force of the submersible and establish the kinematic equations. Step2: To integrate complex ocean parameters into mathematical analysis subtly, we introduce **parameter perturbations** of the stochastic process. At the same time, data is imported to simulate the seafloor topography. Step3: To clarify the parameter disturbance range, we conduct simulation tests. As trajectory of the submersible is greatly affected by uncertainty in the ocean. So, we analyze the factors causing uncertainty to determine the necessary equipment for the submersible to measure these parameters. Finally the information is sent to the main ship to reduce uncertainty.

For question 2, considering costs and benefits, we use EWM and AHP to assign weights to the given equipment. Then, we start from the TOPSIS and Gray Correlation Model based on the entropy weight method. After the importance of different devices is sorted, we analyze the correlation between each device and obtain the comprehensive screening results .

For question 3, we set the position predicted by Model I as the initial search point; then we divide the area of 2 latitudes around the search point into several rectangles. Relied on the results obtained after searching each rectangle, a **Bayesian update** of the prior information is performed to calculate the probability that the submersible is located within the remaining unsearched rectangles. After numerical simulation, we found the search success rate without Bayesian intervention was **76.25%**, while the search success probability increased to **88.69%** after adding Bayesian interve.

For question 4, in order to extrapolate the above model to other tourist destinations, just chooses other ocean parameters and important parameters of manned submersibles, and changes the disturbance factor and range of the model. Finally, we introduce multi-submersible collaborative search, perform **network communication** between submersibles, and establish a multi-agent search and rescue model based on **function mutation theory** to achieve more efficient search and rescue.

Sensitivity analysis shows that the model established in this paper has strong stability under the influence of similar disturbances within the observation tolerance time. When determining the additional equipment required for the main ship and rescue ship, we combined subjective and objective weighting methods to reduce fluctuations and obtained a more comprehensive optimization result.

Ultimately, we designed a two-page safety plan to win the company's approval from the Greek government.

Key words: Random perturbation of parameters; Bayesian update; Function mutation; Network communication;

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1 Introduction

1.1 Background

The underwater world is brimming with mysteries, representing a realm of enigma that humanity has sought to explore. Submersibles serve as the primary means for people to delve into the ocean depths, ferrying them into the deep blue, such as embarking on a journey to "seek the ancient shipwrecks of the Ionian Sea."

However, before initiating any exploration, it is paramount to ensure unwavering stability. The unexpected loss of propulsion, sudden severance of ropes, ocean current impacts, and terrain disturbances are among the myriad "misfortunes from out of the blue" that could lead to severe consequences. Therefore, as responsible engineers at MCM, we are tasked with developing a comprehensive submersible positioning prediction model, selecting the most suitable safety and security instruments, and devising the most thorough search and rescue strategies. Our goal is to construct an impregnable safety barrier for passengers embarking on underwater journeys.

1.2 Problem restatement

Locate: Develop a model for tracking the position changes of an untethered submersible in the ocean that has experienced mechanical and electronic failures. This model needs to consider the effects of natural factors such as ocean currents, water density changes, and seafloor topography.

Uncertainty: Considering the risk of failure and potential position excursions, investigate what information the submersible should send to the host ship during normal operations to reduce uncertainty in post-accident searches, and the type of equipment required to accomplish this task.

Prepare: Evaluate what search equipment the host ship and potential rescue vessels need to carry to effectively locate and rescue the submersible if it loses contact, while taking into account the cost-effectiveness and maneuverability of the equipment.

Search: Develop a search and rescue model based on the submersible position prediction model to guide the selection of the initial search area and search and rescue path, aiming to shorten the time to find the lost submersible and evaluate the probability of finding the submersible at different time points.

Extrapolate: Extend the model to other sea environments and adjust the model to coordinately control multiple submersibles in an area to simultaneously launch search and rescue operations for missing submersibles.

Letter: Write a two-page reporting memo to obtain approval from Greek government regulators for the company's undersea travel business.

1.3 Literature review framework

- By referring to the paper on underwater glider exit position prediction[1], we learned about the six-degree-of-freedom coordinate system of the submersible underwater from a novel perspective, and based on this, we constructed the dynamic equations and kinematic equations in question 1.
- At the same time, we carefully studied a document[2] on the changes in ocean currents in the

Ionian Sea in recent years, and finally established the direction of ocean current movement under ideal conditions.

- For the third question, we studied articles related to Bayesian search[3], and based on their Bayesian update ideas, we built a search model for question three.
- Finally, we learned about the networking communication and detection of underwater multi-robots[4] and unmanned submersibles[5]. Combined with the multi-submersible status mentioned in the fourth question, we considered the collaborative relationship between multiple submersibles. , and improved the search model accordingly.

1.4 Our work

This topic selection requires us to determine a reasonable plan to rescue the lost submersible. Our work mainly includes:

- Based on the flow field and seabed topography of the Ionian Sea, a dynamic prediction model of the submersible based on physical models and random parameter disturbances was established to locate the lost submersible.
- Use probabilistic statistical models to analyze the error and stability of the position prediction model, send information to the main ship to reduce uncertainty, and find professional instruments corresponding to these physical parameters.
- Considering the cost and benefit of carrying equipment, TOPSIS and gray correlation models based on the entropy weight method were established to obtain the importance ranking and correlation coefficient.
- Divide the search area into a state matrix, establish a search model based on Bayesian update theory, and calculate the maximum probability search range.
- Introduce multi-submersible collaborative search, establish a multi-agent search and rescue model based on function mutation theory, and perform network communication to achieve more efficient search and rescue.
- Write a two-page reporting memo to obtain approval from Greek government regulators for the company's undersea travel business.

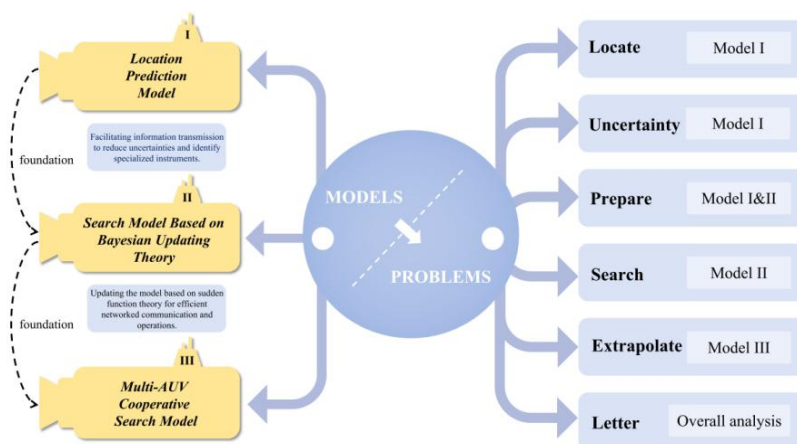


Figure 1: A flow chart of our work

2 Assumption and Justification

In order to simplify the problem, we made the following basic assumptions, each of which is

reasonable. Other assumptions may not be as follows, but will be proposed later in the model.

Assumption 1: Ignore the delay time in receiving information from detection equipment such as sonar.

Justification: The propagation speed of sound waves in seawater is between 1450-1540m/s, which is negligible for the scope of model building we are considering.

Assumption 2: When calculating the problem, treat the submersible as a particle, a cylinder or a related variant of the cylinder (select according to the specific situation).

Justification: This is a common way of simplifying a problem, such as treating an object as a particle in physics

Assumption 3: We believe that the multi-submersible situation in the fourth question means sending multiple submersibles to find the missing submersible.

Justification: The question can be understood in many ways. We assume the situation most relevant to the question positioning and search model.

Assumption 4: The submersible may be on the seafloor or reach a neutral buoyancy point underwater.

Justification: Due to the density of seawater, the buoyancy provided by it is equal to the gravity of a completely submerged object.

3 Notations and Data

3.1 Notations

Table 1 : Declaration of Notations and Signs

Notations	Explanations	Unit
x, y, z	The three-dimensional coordinates of the submersible	m
v_x, v_y, v_z	The three-dimensional speed of the submersible	m/s
m	Submersible quality	kg
F	The force on the submersible	kg·m/s ²
ρ	Sea water density	kg/m ³

3.2 Data collection

Table 2 : Dataset source

Database name	Database Website
Google Earth	https://earth.google.com/web/
GEBCO_ 2023 GRID	https://www.gebco.net/data_and_products/gridded_bathymetry_data/gebco_2023/
NOAA	https://www.noaa.gov/
EARTHDATA	https://www.earthdata.nasa.gov/
Alibaba	https://www.alibaba.com
ADVANCED NAVIGATION	https://www.advancednavigation.com/acoustic-navigation/usbl/subsonus/
Closed circuit diving rebreather	https://www.nauticexpo.com/prod/vobster-marine-systems/product-70296-516230.html

4 Model I: Dynamic prediction model for random parameter perturbations

4.1 Basic idea

For this problem, it is first necessary to clarify its key parameters. we need to conduct a comprehensive collection of environmental parameters in the relevant sea areas.

To simplify the model, we can treat the submersible as a cylinder, which can make the model closer to reality. Because the geographical environment of the seabed will affect the physical parameters in the force analysis, we only need to cleverly integrate this complexity into the mathematical analysis.

The uncertainty in this problem mainly stems from the randomness of the changes in these parameters. Therefore, after completing the model construction, we need to conduct simulation testing to clarify the probability and distribution area. In addition, we need to make measurements of those factors that previously caused uncertainty, send information to the host ship to reduce the uncertainty, and find specialized instruments that correspond to these physical parameters.

4.2 Collection of known information

(1) The manned submersible is about 8 meters long, 3 meters wide and weighs about 3,000 kilograms.

(2) The annual average temperature in the Ionian Sea is 16-19.5°C

(3) Ocean parameters

The daily maximum temperature of the ocean occurs around 15:00, and the minimum temperature occurs around 6:00. The temperature changes relatively slowly, so within the error tolerance, it may be written as:

$$T_0 = \begin{cases} \left| 19.5 \times \frac{t-6}{9} + 16 \times \frac{15-t}{9} \right| & 6 < t < 15 \\ \left| 19.5 \times \frac{t-30}{15} \right| + \left| 16 \times \frac{15-t}{15} \right| & t > 15 \end{cases} \quad (1)$$

The following formula can be used to describe the relationship between seawater temperature and depth:

$$T = T_0 - A \cdot e^{-bh} \quad (2)$$

T is the temperature of sea water, T_0 is the temperature of sea surface, A and b are constant parameters, and h is the depth of sea water[6].

The relationship between seawater density and salinity ($5 - 70\text{g}\cdot\text{kg}^{-1}$) and temperature ($273.15 - 363.15\text{ K}$):

$$\rho = \rho_0 + \alpha(S - S_0) + \beta(T - T_0) \quad (3)$$

ρ is the density of seawater, ρ_0 is the reference density, α and β are the influence coefficients of temperature and salinity on the density of seawater, S is the salinity of seawater, S_0 is the reference salinity, T is the temperature of seawater, and T_0 is the reference temperature.

According to the literature, the change in salinity can be approximated as $S = S_0 - kh$, Comprehensively bring in and sort out:

$$\begin{cases} \rho h = \rho_0 + k_1 h + k_2 e^{-k_3 h} \\ k_1 = k \cdot \alpha, k_2 = A \cdot \beta \\ k_3 = b \end{cases} \quad (4)$$

4.3 Location prediction

- (1) Perform a force analysis on the submersible, and project the resultant forces in the horizontal, vertical, and rotational directions of the submersible.
- (2) Quantify the thrust of the submersible in any period of time based on the law of conservation of momentum.
- (3) The trajectory of the submersible at each moment is quantified by using coordinate equations. Python and Matlab are used to simulate and generate the trajectory of the submersible.

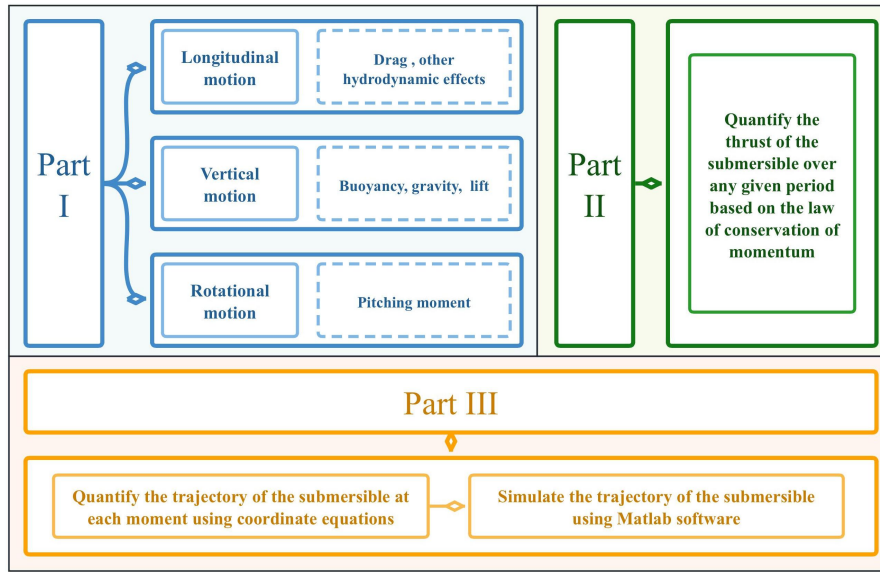


Figure 2: Diagram of the model solving process

a. Determine the position change model of the basic kinematics:

Use a three-dimensional coordinate system to represent the position of submarine.

$x(t), y(t), z(t)$: the position of the submarine at time t .

$v_x(t), v_y(t), v_z(t)$: the speed of submarine in three directions respectively at time t .

Establish the basic kinematic model of submersible based on physical principles.

$$\vec{X}_t(x, y, z) = \vec{X}_0(x_0, y_0, z_0) + \vec{v}_0 t - \frac{1}{2} \vec{a} t^2 \quad (5)$$

Suppose the mass of the submersible is m , the buoyancy force it receives is F_b , the gravity is F_g , the force exerted by the water flow on the submersible is F_c , and the resistance of the submersible in the water other than ocean currents is F_d , then the motion equation of the submersible can be expressed as:

$$m \frac{d^2 \vec{r}}{dt^2} = \vec{F}_b + \vec{F}_g + \vec{F}_c + \vec{F}_d = \vec{F}_A \quad (6)$$

According to the momentum theorem when $t=t+\Delta t$,

$$(m\vec{v}_{t_i-\Delta t} - m\vec{v}_0) = \sum_{i=1}^n \vec{F}_{A_i} \Delta t \quad (7)$$

Integrate time Δt ,

$$\int_{\Delta t=0}^t (m\vec{v}_{t_i-\Delta t} - m\vec{v}_0) = \int_{\Delta t=0}^t \sum_{i=1}^n \vec{F}_{A_i} \Delta t \quad (8)$$

b. Introduce uncertainty and determine the range of parameter disturbance:

To predict the position of the submersible, we build a dynamic model adding random parameter disturbance based on the kinematic model.

We need to determine the range of parameter perturbations. These perturbation ranges can be determined from experimental data, literature studies, or expert experience.

We first consider a simple perturbation model: ocean density perturbation. Assuming that changes in ocean density have an impact on the motion of the submersible, the following equation can be used to describe ocean density disturbances:

$$\rho(x, y, z, t) = \rho_0 + \delta\rho(x, y, z, t) \quad (9)$$

Among them, ρ_0 is the basic density of the ocean, and $\delta\rho$ is the perturbation term of the ocean density. A stochastic process is introduced to represent the uncertainty of the external environment.

$$\begin{cases} \frac{dx}{dt} = f_x(x, y, z, t) + \delta_x(t) \\ \frac{dy}{dt} = f_y(x, y, z, t) + \delta_y(t) \\ \frac{dz}{dt} = f_z(x, y, z, t) + \delta_z(t) \end{cases} \quad (10)$$

c. Generate perturbation parameters: After determining the perturbation range, you can use random number generation methods to generate perturbation parameters.

d. Run multiple predictions: Use the basic kinematic model and disturbance parameters to perform multiple position predictions. Each prediction uses different perturbation parameters to simulate the uncertainty in the real environment.

e. Analysis results: By analyzing the results of multiple predictions, statistics such as average position, standard deviation or confidence interval. These statistics can provide information about the accuracy and reliability of the submersible's position predictions.

4.3.1 Communication failure hypothesis

Assume that the submersible loses communication with the host ship, but still has propulsion and is at a neutral buoyancy point.

$$m \frac{d^2 \vec{r}}{dt^2} = \vec{F}_b + \vec{F}_g + \vec{F}_c + \vec{F}_d = \vec{F}_A \quad (11)$$

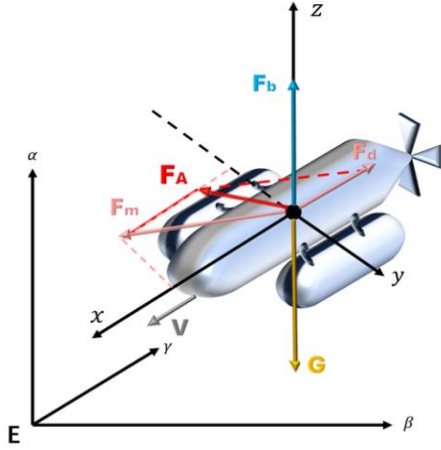


Figure 3: Force analysis diagram

While the submersible is propelled by force, it also experiences resistance from the water. To move forward or maintain position, the submersible must overcome water resistance, which is affected by shape and speed. As speed increases, resistance increases accordingly. Here, a randomly generated method is used to simulate the resultant force, which is adjusted to a normal distribution based on the original acceleration and ensures that the acceleration is non-negative.

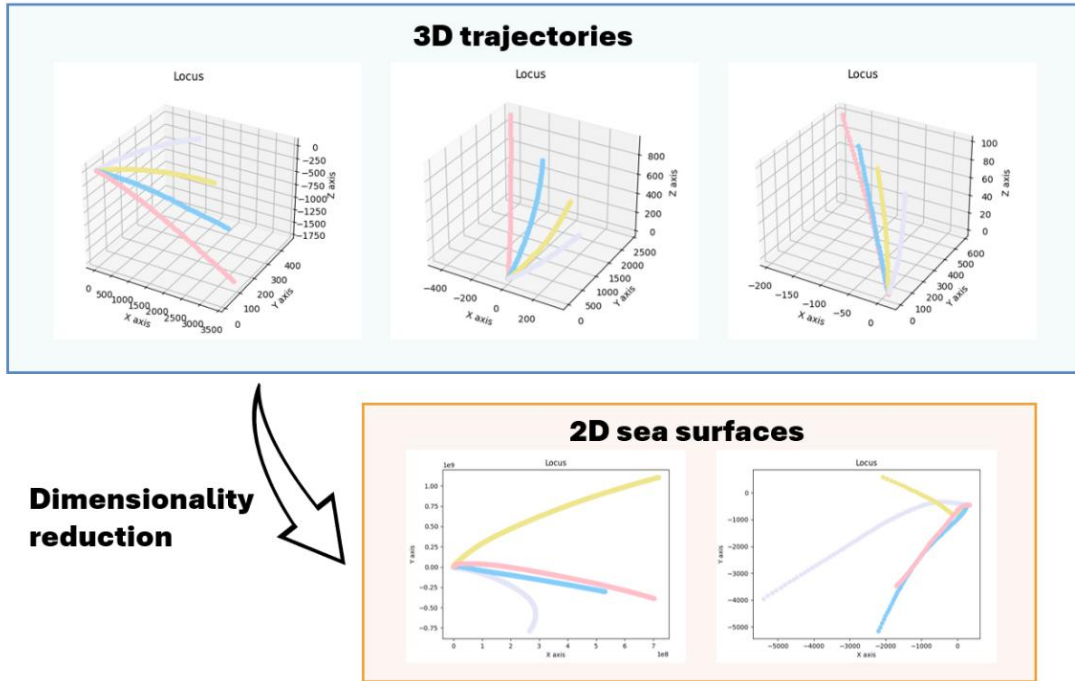


Figure 4: Trajectory diagram with flow field disturbances

4.3.2 Ocean current disturbance hypothesis

Assume that the submersible is at a neutral buoyancy point in the water and loses propulsion. The stationary Navier-Stokes equation under stationary conditions is as follows,

$$\frac{\partial}{\partial t} \rho \vec{u} + \vec{u} \cdot \nabla \rho \vec{u} = \rho \vec{F} - \nabla p + \mu \nabla^2 \rho \vec{u} \quad (12)$$

For the side, we can simplify it to a typical flow model around a cylinder. We can think of the problem as a incompressible flow passing through a cylinder, which may oscillate laterally due to the lift exerted by the fluid. The filtered two-dimensional Navier-Stokes equation is represented by the Poisson equation of the stream function:

$$\nabla^2 \psi = -\omega \quad (13)$$

Vorticity transport equation:

$$\frac{\partial \omega}{\partial t} + u \frac{\partial \omega}{\partial r} + \frac{v}{r} \frac{\partial \omega}{\partial \theta} = (v + v_t) \left(\frac{\partial^2}{\partial r^2} + \frac{1}{r} \frac{\partial}{\partial r} + \frac{1}{r^2} \frac{\partial^2}{\partial \theta^2} \right) \omega \quad (14)$$

For the front, we might as well simplify it to a capsule-like object:

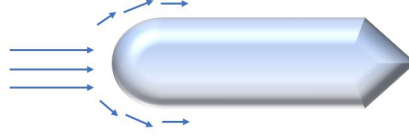
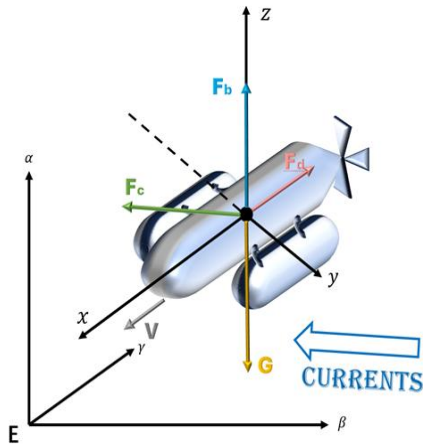


Figure 5: Abstract drawing of submersible shape

Existing literature provides a continuous function that is valid for all distances from the surface, a drag force interpolation model for capsule-shaped cells in fluid flows near a surface. Fitting a function is as follows:

$$f\left(\frac{H}{R_e}\right) = 0.700 \left(\frac{H}{R_e}\right)^{-1.082} + 1.001 \quad (15)$$

So far we have obtained the expression of the force that the ocean current affects the submersible from the front and side.



Suppose the force exerted by the ocean current on the submersible is F_c , and the resistance of the submersible in the water other than ocean currents (=hydrodynamic resistance) is F_d , then the submersible's The equation of motion can be expressed as:

$$m \frac{d^2 \vec{r}}{dt^2} = \vec{F}_b + \vec{F}_g + \vec{F}_c + \vec{F}_d \quad (16)$$

Figure 6: Force analysis diagram

The average monthly ocean state of the Ionian and Adriatic Seas over the 60-year period 1955-2015 is explored by reanalyzing existing data sets, exploiting vorticity fluxes based on reconstructed monthly average surface velocity fields. The Ionian circulation is analyzed, which always flowing northward along the Greek Trench and southward along the Italian coastline.

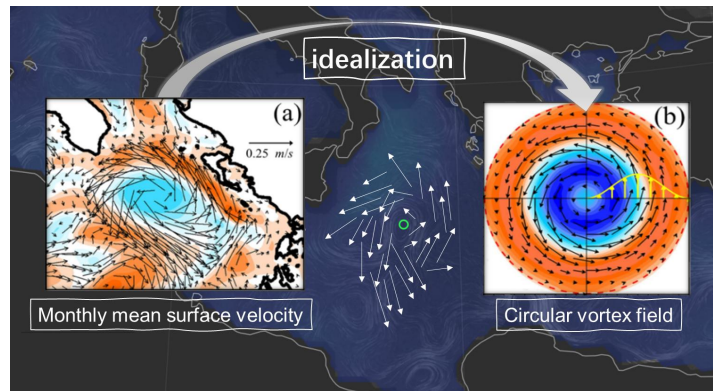


Figure 7: Idealization of the ocean flow field

In this question, we set the ocean current speed to [0.257, 0.771] (m/s), and use the ocean current center (20°E, 38 °N) and the radial direction of the radius with its center as the acceleration direction.

If the time change is extremely small, the limit method can be used to calculate the force exerted by the ocean current on the submersible. At this time, the force of the ocean current on the submersible is approximately constant. According to the momentum theorem, force is the product of mass and the rate of change of velocity.

Assume that the initial speed is V_t , the mass of the ocean current is ignored, and the speed is V_c . According to the momentum theorem, the force F exerted by ocean currents on the submersible can be expressed as:

$$\vec{v}_{rela} = \vec{v}_t - \vec{v}_c \quad (17)$$

$$\vec{F}_c = m \cdot \frac{\vec{v}_{rela}}{\Delta t} \quad (18)$$

When a submersible moves in seawater, it will encounter resistance, which is related to the density of seawater. Resistance can be divided into friction resistance and pressure difference resistance. This can be approximately calculated by the following formula:

$$\vec{F}_d = \frac{1}{2} \rho_t C_d A \vec{v}_{rela}^2 \quad (19)$$

V_{real} is the speed of the submersible relative to seawater (m/s), C_d is the resistance coefficient, A is the water facing area of the submersible (cubic meters, m^2)

Since the search work here is often measured in hours, the model here should be based on minutes. Minutes is too large a span for the sinking itself. Use One state is coupled to the previous state :

$$\begin{cases} x_n = x_{n-1} + v_{x_{n-1}} dt \\ y_n = y_{n-1} + v_{y_{n-1}} dt \\ z_n = z_{n-1} + v_{z_{n-1}} dt \end{cases} \quad (20)$$

Bringing in the meta-analytic expression we get:

$$\begin{cases} a_x = \frac{d^2x}{dt^2} = f\left(\frac{H}{R_e}\right) \cdot \frac{6\pi\eta R_e}{m} \cdot X V_x \\ a_y = \frac{d^2y}{dt^2} = \frac{1}{2} \cdot \frac{C_d \rho (Y V_y)^2 A}{m} \\ a_z = \frac{d^2z}{dt^2} = \frac{(\rho_0 + k_1 z + k_2 e^{-k_3 z}) g V}{m} - g - uv \end{cases} \quad (21)$$

X and Y change after each time step. The following shows some random XY coordinate drift trajectories.

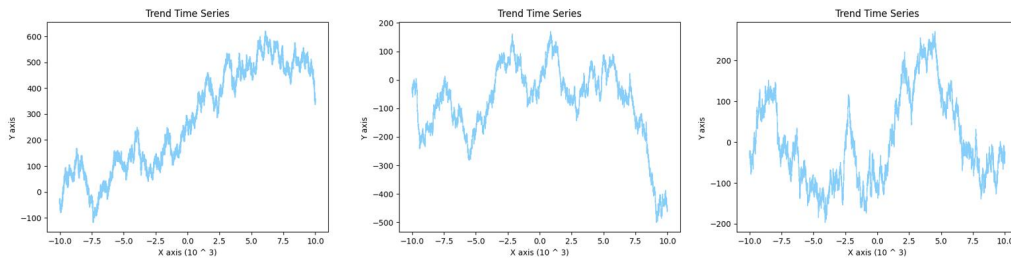


Figure 8 : Some random XY coordinate drifting trajectories

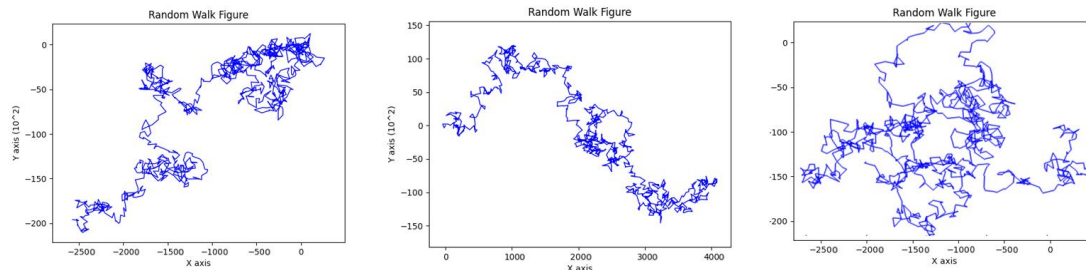


Figure 9 : XY coordinate drifting trajectories with changes in all directions

4.3.3 Terrain Disturbance Hypothesis

Suppose the submersible is closer to the seafloor and loses propulsion. If the submersible is relatively close to the seafloor, the terrain disturbance to the direction of the submersible cannot be ignored and the terrain disturbance factor must be added.

After investigation, it was found that the Ionian seafloor topography is dominated by monoclinic continental slope belt topography and erosion depression topography.

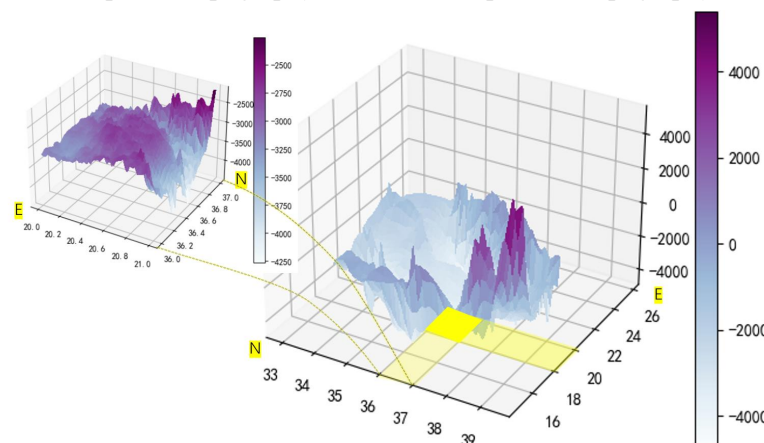


Figure 10 : 3D map of topographic data

When a submersible traverses a monoclinic continental slope terrain, due to the wall effect, the pressure on one side of the slope is greater than the other side, and the seawater will exert a force perpendicular to the slope on the surface of the submersible, which is called "pressure resistance", and the speed will slow down due to friction.

In depressed areas of the ocean floor, the flow of water may slow down. Depressions are low-lying areas of land where surrounding water flows tend to converge. As the flow rate increases, the velocity of the water flow may decrease relatively.

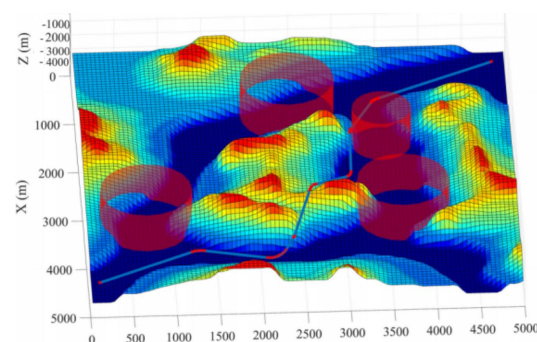


Figure11: Three-dimensional path

We assume that it still has good sensors and can perform directional control through the rudder surface, so here we introduce an obstacle avoidance algorithm based on genetic algorithms. The submersible continuously interacts with the environment through its own sensors, thereby obtaining previously unknown environmental information. On this basis, we can establish a three-dimensional path model of the near-seabed submersible on the seafloor map.

4.4 Uncertainty Analysis that Hinders Forecasting

There are many uncertainties in predictive models, such as possible errors or noise in the data itself, or the inability to fully represent the entire system to be simulated. Missing data, errors, and deficiencies in the prediction model may significantly lead to uncertainty. Taking our problem as an example, we cannot obtain an accurate 3D seafloor topographic map.

In addition to data factors, blindly simplifying the actual factors of the hypothetical model will also increase the uncertainty of the model. For example, we have simplified natural environmental factors such as the direction of ocean currents on the seabed and the density of seawater, which will lead to an increase in uncertainty.

At the same time, the parameters of the hypothetical model are also one of the reasons for the birth of uncertainty. During the process of model building, it may be difficult for us to obtain exact parameter data. Many of the natural data, such as seafloor water pressure, water temperature, and topography, may need to rely on estimates of historical data, and therefore inevitably contain estimation errors.

Finally, considering that the current development of human science and technology has not yet reached a super advanced level, even if we have equipment to obtain actual data, we have to consider limitations of scientific and technological conditions such as sensor accuracy, communication delay, and exploration range, and uncertainty is naturally unavoidable .

4.5 Ways to reduce uncertainty

4.5.1 Send relevant information regularly

Regularly sending information about the submersible can help reduce uncertainty, such as location information, which can improve the accuracy of predictive models.

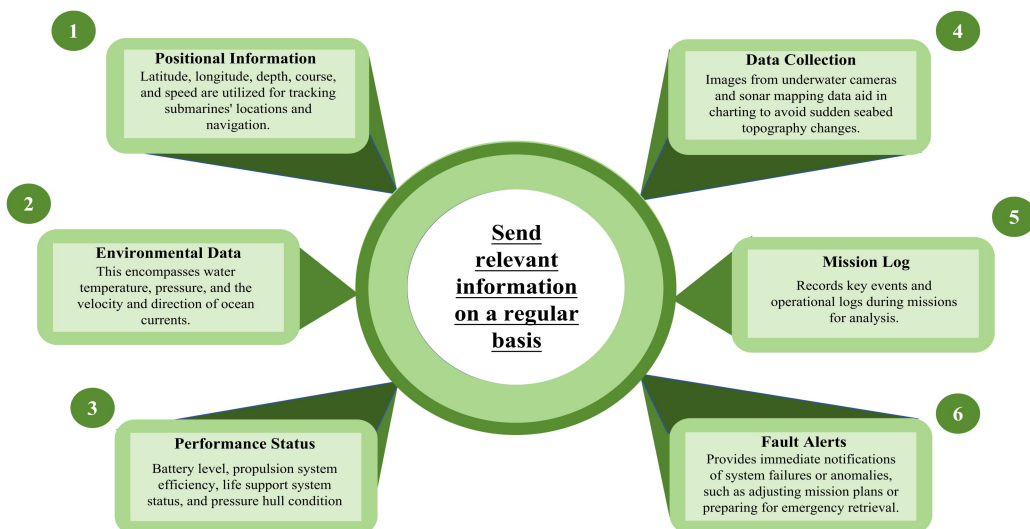


Figure 12: Ways to reduce uncertainty

4.5.2 Send relevant information regularly

For the relevant information that is transmitted regularly, we selected a series of matching information transmission devices and displayed them visually.

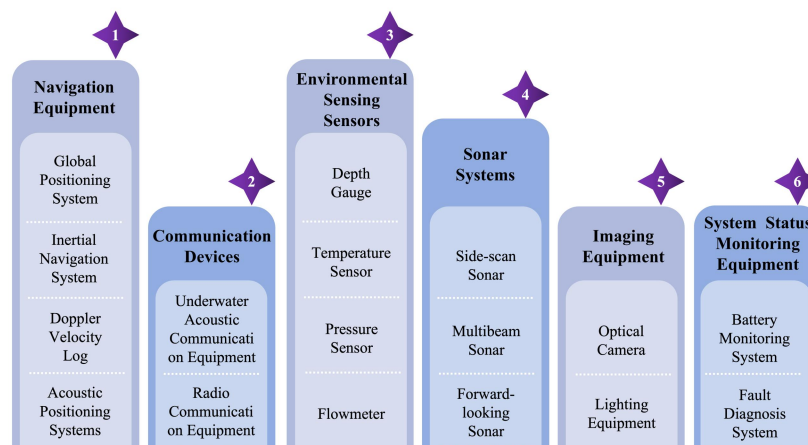


Figure 13: Equipment to send information

5 Model II: Recommended search equipment to carry

5.1 Equipment carried by the main ship

5.1.1 Preparation

After Internet search and investigation, we identified some of the most common marine search and rescue equipment all of them are listed below.

When evaluating and analyzing the main ship equipment, we invited experts to rate its relative usability (1-10) and searched the preparation costs of all equipment on various online platforms.

- The price of life-saving equipment is 5 sets, and the maintenance cost and usage cost are the price for one year under normal use.
- Since the maintenance cost of various equipment is affected by multiple factors like the frequency, operating environment and technical complexity. In order to obtain relevant data more conveniently, we stipulate all factors related to the use of the equipment. All are under normal conditions, and according to convention, the maintenance cost is limited to 3% - 10% of the preparation cost, and the use cost is 6% - 9% of the preparation cost, and the following data table is obtained:

Table3: Table of device metrics

	Availabi lity	Maintenance/ \$	Readiness/ \$	Usage/ \$	Summary/ \$
Voice-wrenching	8	15641	173792	1240	190673
Stallite communication equipment	6	890	22245	487	23622
Meteorological observers	7	1112	13903	1251	16266
Life-saving equipment	9	52	1738	104	1894
Fully enclosed scuba diving respirator	9	411	5868	411	6690
Acoustic positioning system	8	1127	11861	1067	14055
Emergency beacon receiver	6	9	240	19	268
AUV	10	17100	180000	14400	211500
ROV	10	2113	22245	1180	25538

Construct the data in the table into a data matrix, and perform forwarding and standardization operations on it to obtain the following matrix:

$$\begin{bmatrix} 8 & 1459 & 173552 & 1121 \\ 6 & 16210 & 22005 & 468 \\ 7 & 15988 & 13663 & 1232 \\ 9 & 17048 & 1498 & 85 \\ 9 & 16689 & 5628 & 392 \\ 8 & 15973 & 11621 & 1048 \\ 6 & 17091 & 0 & 0 \\ 10 & 0 & 179760 & 14381 \\ 10 & 14987 & 22005 & 1161 \end{bmatrix}$$

(a) Forwarding Matrix

$$\begin{bmatrix} 0.3236 & 0.0338 & 0.6873 & 0.0837 \\ 0.2427 & 0.3757 & 0.0871 & 0.0321 \\ 0.2832 & 0.3706 & 0.0541 & 0.0845 \\ 0.3641 & 0.3951 & 0.0059 & 0.0058 \\ 0.3641 & 0.3868 & 0.0223 & 0.0269 \\ 0.3236 & 0.3702 & 0.0460 & 0.0719 \\ 0.2427 & 0.3961 & 0 & 0 \\ 0.4046 & 0 & 0.7119 & 0.9862 \\ 0.4646 & 0.3474 & 0.0871 & 0.0796 \end{bmatrix}$$

(b) Normalized Matrix

5.1.2 TOPSIS method based on entropy weight method

First, use the entropy weight method to set the weight of the preprocessed matrix. After running the code in Matlab, the weights of the four indicators can be obtained:

$$0.0074, 0.0944, 0.3968, 0.5014$$

Then we continued to use Matlab to complete the solution for the total score, and sorted the 9 devices according to their scores as follows:

$$8, 1, 9, 3, 6, 2, 5, 4, 7$$

However, due to the limitations of data acquisition, the weights determined by the entropy weight method are not completely certain. Therefore, we plan to conduct the second type of evaluation - gray correlation analysis to verify the correctness of the equipment ranking.

5.1.3 Gray correlation method

Since we have previously obtained the normalized matrix of the data, we performed gray correlation analysis operations by Python to obtain its correlation coefficient diagram.

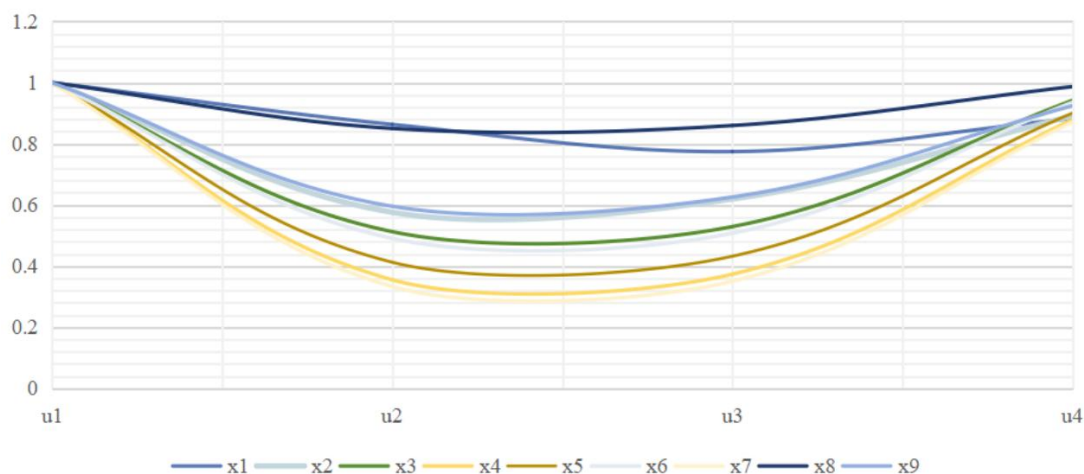


Figure 14: Correlation coefficient plot

Next, we determined the weights of relevant indicators through the correlation coefficient, and calculated the scores of all devices based on the weights. After normalizing the scores, we finally obtained the ranking of the 9 devices as follows:

$$8, 1, 9, 2, 3, 6, 5, 4, 7$$

We can find that the final results of TOPSIS and gray correlation analysis are mostly the same, but there is still a ranking shift of device 2, so we list the two categories of scores in a table for comparison:

Table4: Comparison table of review scores

Xi	TOPSIS Score	GRA Score
1	0.1291	0.0999
2	0.1075	0.0916
3	0.1132	0.0916
4	0.0956	0.211
5	0.1006	0.0674
6	0.1094	0.0919
7	0.0937	0.0903
8	0.1355	0.1605
9	0.1153	0.0959

5.1.4 Comprehensive screening

According to the function and use of the equipment, we can divide the equipment into the following three categories:

Communication and positioning equipment: satellite communication equipment, acoustic positioning system, emergency beacon receiver.

Observation and detection equipment: sonar, meteorological observers, AUV, ROV.

Rescue and survival equipment: life-saving facilities, fully enclosed diving breathing apparatus.

In order to improve the efficiency and reliability of search and rescue, we selected a set of the most effective equipment from all equipment, which need to meet:

- ◆ Select the devices in order according to the scores given previously.
- ◆ Make sure at least one device in each category is selected

Set the weights of TOPSIS and gray relational analysis to 0.5, combine the listed constraints, and program. Finally, we get the selected equipment as acoustic positioning system, AUV (unmanned submersible), and fully enclosed diving respirator.

5.2 Equipment carried by rescue boats

For the additional equipment carried by the rescue ship, we also performed the above operations, and finally obtained the portable equipment: diving support equipment, emergency rescue tools and equipment, navigation and control systems.

6 Model III :Search model based on Bayesian updating

theory

The search model based on the probability map combines the submersible position prediction model, and uses Bayesian Search Theory to determine the best search area and search path, minimizing the search time to connect the submersible.

To utilize the position model, we use the predicted target position as the integration center. Based on the uncertainty of the prediction, we roughly divide several areas and correct them into regular rectangles.

The search area is divided into multiple grids, and probability P_i of each grid i represents the probability that the submersible exists in the grid. The search efficiency η_i depends on the type of

search device and the search environment.

Assume that the last information sent by the searched submersible to the mother ship is its latitude, longitude and depth at that time. Combined with the equipment selected in the second question, we consider sending the AUV to start a head-up search at the depth sent by the submersible, carrying the acoustic positioning system and mainly pointing up and down to Explore in depth.

6.1 Model Building

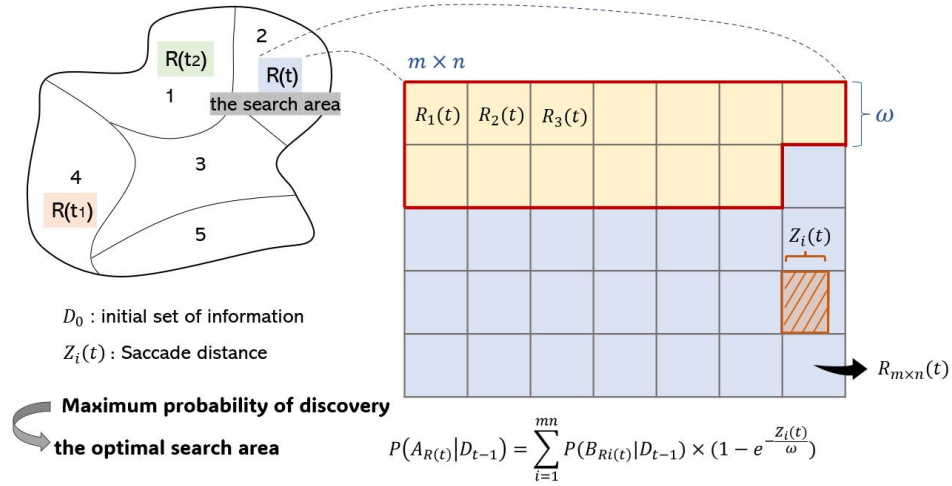


Figure 15: Prediction map of sea search and rescue

The initial information set is recorded as D_0 , and the event that the target is located in the R_k area is recorded as B_{R_k} . Based on this, the prior probability of $P(B_{R_k} | D_0)$ is established.

Let $R(t)$ be the search area in the t -th stage, consisting of $m \times n$ connected small squares $R_1(t), R_2(t), \dots, R_{m \times n}(t)$ in $R_1, R_2, \dots, R_N$. so the search area in the t -th stage is mnw^2 ,

The probability that the drone searches and finds the target in the area $R(t)$ is recorded as $A_{R(t)}$, and its probability is

$$P(A_{R(t)} | D_{t-1}) = \sum_{i=1}^{mn} P(B_{R_i(t)} | D_{t-1}) P(A_{R_i(t)} | B_{R_i(t)}, D_{t-1}) \quad (22)$$

The probability of finding the target when the target is located at $R_i(t)$ is

$$P(A_{R_i(t)} | B_{R_i(t)}, D_{t-1}) = 1 - e^{-\frac{Z_i(t)}{w}} \quad (23)$$

Record the scanned distance of each small area $R_i(t)$ in $R(t)$ as $Z_i(t)$, and combine the last two formula, we can get

$$P(A_{R(t)} | D_{t-1}) = \sum_{i=1}^{mn} (1 - e^{-\frac{Z_i(t)}{w}}) P(B_{R_i(t)} | D_{t-1}) \quad (24)$$

We use the discovery probability to determine the optimal search area. In principle, the $R(t)$ with the highest probability is taken as the search area. If the target has not been found when stage t is reached, we perform a Bayesian update on the prior probability and start the search in stage $t+1$:

$$D_t = \{\bar{A}_{R(t)}, D_{t-1}\}$$

Then we can get the posterior probability that the target is located in the area R_k at stage $t+1$:

$$P(B_{R_k} | D_t) = \frac{P(B_{R_k} | D_{t-1}) P(\bar{A}_{R(t)} | B_{R_k}, D_{t-1})}{\sum_{R_j \in R(t)} [P(B_{R_j} | D_{t-1}) P(\bar{A}_{R(t)} | B_{R_j}, D_{t-1})] + \sum_{R_j \notin R(t)} P(B_{R_j} | D_{t-1})} \quad (25)$$

$$= \frac{e^{-\frac{Z_k(t)}{\omega}} P(B_{R_k} | D_{t-1})}{\sum_{R_j \in R(t)} e^{-\frac{Z_j(t)}{\omega}} P(B_{R_j} | D_{t-1}) + \sum_{R_j \notin R(t)} P(B_{R_j} | D_{t-1})}$$

$R(t+1)$ is the search area in the $t+1$ stage. Combine it to the last two formula, we can get the posterior probability $P(A_{R(t+1)} | D_t)$.

Therefore, after the search proceeds for s stages, we can get the finally P :

$$P = 1 - \prod_{t=1}^s [1 - P(A_{R(t)} | D_{t-1})] \quad (26)$$

6.2 Optimal Search Strategy

To find the strategy that maximizes probability under given conditions, the scanning speed per unit time is v_0 , and the maximum event in a search is T .

The optimal search strategy is planned as follows:

$$(Z^*(t), R^*(t)) = \arg \max_{Z(t), R(t)} \sum_{i=1}^{mn} \left(1 - e^{-\frac{Z_i(t)}{\omega}}\right) P(B_{R_i(t)} | D_{t-1}) \quad (27)$$

$$\text{s.t. } z_{i(t)} \geq \omega, i = 1, 2, \dots, mn, \sum_{i=1}^{mn} z_{i(t)} \leq \gamma_0 T, R(t) \subset \langle R_1, R_2, \dots, R_N \rangle \quad (28)$$

6.3 Numerical Simulation

Let the number of base points (the base point is the most likely location of the target) be recorded as M , and the distribution probability of the target located in the area where the k th base point is located be recorded as $f_k(x, y)$, let $f_k(x, y)$ be a two-dimensional normal distributed.

Assume that the last known position of the submersible is the mean. We can estimate that the submersible may move up to 2 units of latitude and longitude per day. Therefore, we can set the standard deviation $\delta_a = 2.0$, $\delta_y = 2.0$. Assuming relative independence between longitude and latitude variations, the correlation coefficient is $\rho = 0$.

Based on the mean and standard deviation, we can choose the area around the mean, such as $\mu_x \pm 3\delta_x$ and $\mu_y \pm 3\delta_y$ as the integration range.

As new information becomes available, such as the discovery of parts of a submersible or new environmental data, we need to update the mean and standard deviation.

6.3.1 Simulation experiments without Bayesian intervention

The initial probability calculation formula for the target to be located in the area R_k is as follows:

$$P_{R_k} = \int \int_{(x,y) \in R_k} f_k(x, y) dx dy, k = 1, 2, \dots, N \quad (29)$$

Assume that there are 5 base points A, B, C, D, and E in the Ionian Sea. These 5 base points

divide the ocean into 5 areas, and each area is divided into 3×3 small areas of equal area. The probability distribution and probability information of the five regions are shown in the table below.

Table5: Probability information of the five regions

Area	Probability distributions	Probability
A	$N_A(17, 37; 2, 2; 0)$	$P_A=0.25$
B	$N_B(20, 36; 2, 2; 0)$	$P_B=0.08$
C	$N_C(17, 35; 2, 2; 0)$	$P_C=0.97$
D	$N_D(19, 36; 2, 2; 0)$	$P_D=0.13$
E	$N_E(18, 37; 2, 2; 0)$	$P_E=0.004$

Calculate the probability of each area under the probability distribution of the area, and then select the three areas with the highest probability as the task area. Obviously, we select three areas C, A, and D as task areas R_1 , R_2 , and R_3 respectively. Then the probability of the task area is normalized, and the formula is:

$$P(B_{R_k} | D_0) = \frac{P_{R_k}}{\sum_{k=1}^n P_{R_k}} \quad (30)$$

It can be obtained that, take the scan width as $w = 10$, and the maximum dwell time of each stage $T = 20$. We set $v_0 = \alpha mnw/T$, α is the scanning area coefficient, the larger α is, the greater the work intensity of the drone search equipment in a search stage.

Take α in $[1, 2]$, and the step size is 0.2 for numerical simulation. By solving the optimal search strategy model, the discovery probability of each stage can be obtained as follows:

Table 6: Discovery probability without Bayesian intervention

	$P(B_{R1} D_0)$	$P(B_{R2} D_0)$	$P(B_{R3} D_0)$	P
$a = 1.0$	0.4587	0.267	0.3287	0.6337
$a = 1.2$	0.6576	0.333	0.3844	0.7594
$a = 1.4$	0.6854	0.3546	0.4173	0.7817
$a = 1.6$	0.699	0.3699	0.4405	0.7939
$a = 1.8$	0.7048	0.3813	0.4577	0.8009
$a = 2.0$	0.7089	0.3868	0.4703	0.8055

6.3.2 Simulation experiments with Bayesian intervention

Add Bayesian intervention to the numerical simulation, that is, after completing the first stage of search, add a new information $N_F(6, 13; 8, 8, 0)$, $P_F=0.52 > P_A$, then update the F area to the second Stage search area, area A is the third stage search area. At this time, the discovery probability of each stage is as follows:

Table 7: Discovery probability with Bayesian intervention

	$P(B_{R1} D_0)$	$P(B_{R2} D_0)$	$P(B_{R3} D_0)$	P
$a = 1.0$	0.4587	0.3052	0.3919	0.7713
$a = 1.2$	0.6576	0.3805	0.4502	0.8834
$a = 1.4$	0.6854	0.4052	0.485	0.9036
$a = 1.6$	0.699	0.4227	0.5098	0.9148
$a = 1.8$	0.7048	0.4357	0.5298	0.9217
$a = 2.0$	0.7089	0.442	0.5462	0.9263

As can be seen from the two tables above, after adding Bayesian intervention, the discovery

probabilities increased in last two stages. This is due to the fact that after Bayesian intervention is added, the probability distribution is readjusted. The newly added area after intervention has a higher probability of target existence than the original area, so the discovery probability also increases, and the cumulative discovery probability also increases. The figure below gives a comparison of the cumulative discovery probability with and without Bayesian intervention.

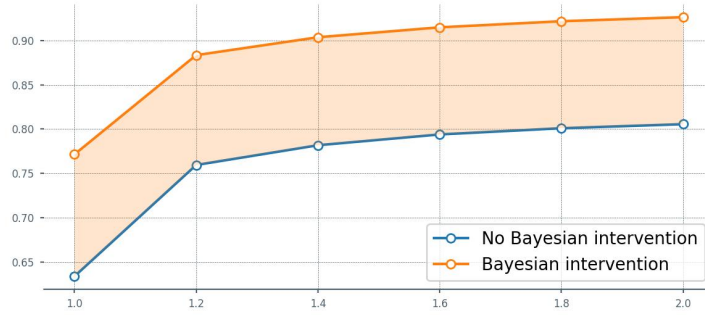


Figure 16: Comparison chart of experiments

7 A Multi-agent model based on catastrophe theory

7.1 Adaptive extrapolation

The actual seawater temperature distribution may be affected by a variety of complex factors, such as geographical location, season, ocean currents, seafloor topography, etc. If more detailed data and background information are available, model parameters can be adjusted for specific situations or a more complex model can be used to more accurately describe seawater temperature as a function of depth.

In the model adjustment, we mainly consider modifying the parameters of $f_k(x,y)$ and increasing the tail of the probability distribution of the base point in the direction of the ocean current, so as to extend the model to other tourist areas.

7.2 Multi-submersible collaborative search

After introducing a multi-submersible cluster formation for collaborative search and rescue, the submersible clusters can communicate with each other, and the same submersible cluster can also communicate with the main ship.

For the seabed plane, it can be divided according to longitude and latitude, and can be divided into $N*M$ different square areas, and each area is numbered. If area i has not been searched, then $Z_i=1$, otherwise $Z_i=0$. Each time the submersible completes searching a square area, a signal is sent to the main ship to change Z_i . When the main ship receives the signal, the state matrix Z will be updated in time, and the state matrix will be put into the prior information D and broadcast to all submersibles.

When the submersible performs Bayesian search and update, if the state matrix is 0, it skips the repeated search and no longer considers it. Since multiple submersibles are dispatched for search and rescue, in order to improve search efficiency, we introduce event C to adjust the search model[7].

When $C=1$, it means that at least one submersible has found the target. $C=0$ means that no

submersible has found the target. When extended to the t -th stage, it is C_t . $C_t=1$ means that the target has been found at this stage and there is no need to proceed to the next stage; $C_t=0$ means that the target has not been found at this stage and the next stage of search is required.

In order to simplify the model, we consider setting up an indicator function I_c . When $C_t=1$, $I_c=1$, otherwise $I_c=0$.

$$P = I_c + (1 - I_c) \cdot \left\{ 1 - \prod_{t=1}^s [1 - P(A_{R(t)}|D_{t-1})] \right\} \quad (31)$$

Therefore, by taking advantage of the mutation nature of the function, we can search better. Finally, the search and rescue process is shown in the following cartoon:

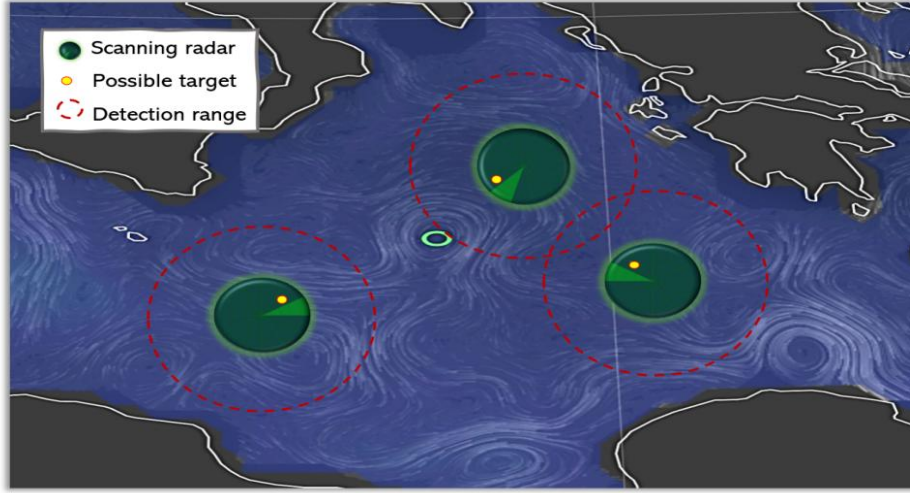


Figure 17: Ways to reduce uncertainty

8 Model Analysis

8.1 Sensitivity Analysis

(1) Regarding the first question, the mass and volume of the submersible, the acceleration due to gravity, and the water-facing area of the submersible are all easily measured and determined. However, the ocean parameters such as seawater density, seafloor flow field, and seafloor topography where the submersible is located have problems such as insufficient observation conditions and difficulty in data acquisition. Therefore, we need to evaluate whether the submersible positioning model is sensitive to disturbance changes in these ocean parameters.

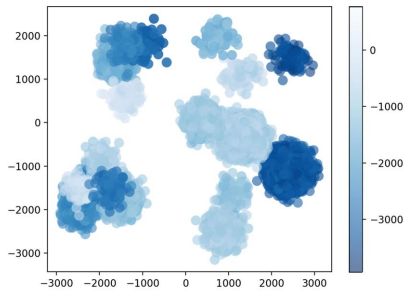


Figure 18: Probability distribution plots

The starting point is set to $(0, 0, -2000)$, the loss of contact time is set to half an hour. By changing the variation range of the disturbance parameters, 10 predicted trajectories are generated in each round, and the position coordinates are respectively Find the mean, standard deviation and confidence interval on the x, y, and z axes.

Table 7 : Probabilistic model analysis

	Average Endpoint			Standard Deviation			Confidence Interval (95%)		
	x	y	x	x	y	z	x	y	z
1	2002.67	-1687.09	-1716.14	184.77	220.75	125.11	[2141.99,1863.34]	[-1853.55,-1520.63]	[-1810.48,-1621.81]
2	2234.52	-888.07	-3753.79	121.17	159.02	126.07	[2143.15,2325.89]	[-1007.98,-768.16]	[-3848.85,-3658.7]
3	923.29	-1547.64	-2846.82	182.04	130.26	182.9	[786.02,1060.56]	[-1645.86,-1449.42]	[-2984.73,-2708.91]

It can be seen from the average value that the changes in ocean parameters in each round have a greater impact on the positioning end point. It can be seen from the standard deviation that the fluctuations in the three dimensions of each round of prediction results are small. When the ocean parameters are the same, the prediction results are similar.

(2) Regarding the third question, due to the availability of search equipment, maintenance costs cannot be accurately given, and factors such as equipment price and cost are subject to large market fluctuations, so we need to conduct stable testing of equipment preparation. We use different methods to analyze the correlation and rank the importance of the additional equipment that the main ship and the rescue ship need to carry. we compare the ranking changes between the two solutions, evaluate the sensitivity of the model to indicator changes, and use a line chart to display the results more intuitively.

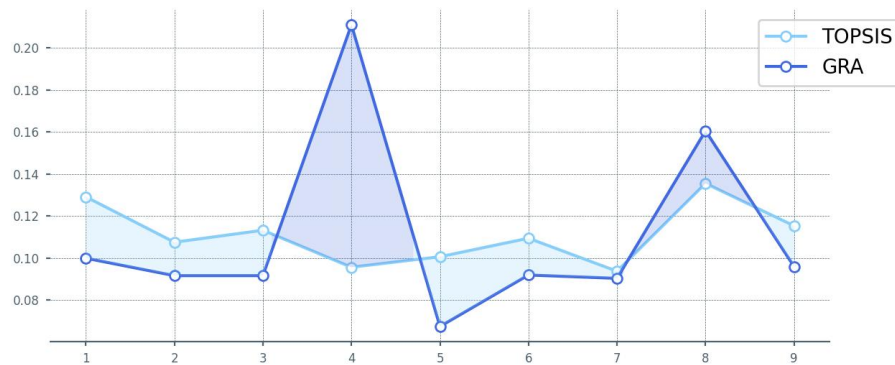


Figure 19: Score comparison chart

8.2 Strength and Weakness

8.2.1 Strengths

- Our model is comprehensive, integrating ocean currents, density, terrain and other factors given in the question, and constructing a series of closely related and effective positioning, search, and collaboration models.
- Our model is very flexible and can be used in different sea areas for underwater submersible detection. For example, you only need to adjust the ocean parameters in the model to switch sea areas; to consider more ocean factors, just add ocean parameters.
- We combine Newton's second law, Bayesian search theory, and network communication methods of drones and submersibles to build a model, which has a strong and reliable theoretical basis.
- In the existing literature, it is mostly assumed that the distribution of targets in the search area is uniform. However, usually, since the distribution of the target area and the geographical feature information of the area are obtained in advance, it is easy to infer that the target is located at certain base points. The distribution probability may be larger. Therefore, the

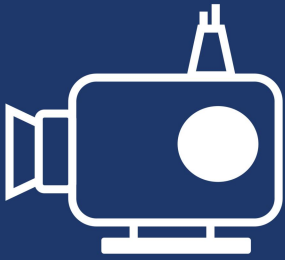
two-dimensional normal distribution assumption adopted by our model appears more reasonable.

8.2.2 Weaknesses

- In making physical assumptions, we simplified the form of the force exerted by the fluid on the submersible, which may cause uncertainty in the force analysis of the submersible, and the range of disturbance parameters has a significant impact on the results.
- In the process of searching for the cost of equipment, due to various factors, we did not directly determine its use cost and maintenance cost, but set up an interval range, which may lead to deviations in our evaluation results.

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<https://doi.org/10.1038/051342a0>



MEMO

Dear Greek government,

We are honored to present our analysis on the issue of predicting the locations of underwater vehicles after comprehensive data analysis and modeling. We are confident that our intriguing findings will prove to be of great value to you. Below, we offer a summary of our theoretical analyses and numerical predictions.

Firstly, we conducted a force analysis on the submersible based on Newton's second law and established a three-dimensional coordinate system for predicting the location of the submersible. Recognizing the complexity of incorporating all factors into the model, we simplified the preconditions before formally establishing the model. For example, we treated the submersible under specific conditions as a point mass, a cylindrical shape or a variant thereof, and idealized the direction of ocean currents. After determining the focal point for model establishment, we gradually incorporated marine parameters such as ocean currents, seawater density, and seabed topography into the model, adding random perturbations to make the model more realistic. Using various programming software, we solved a series of two-dimensional and three-dimensional predictive route maps. In the process, we uncovered several uncertainties, such as **potential errors or noise in the data itself, or the inability to fully represent the system intended for simulation; oversimplification of the model's real-world factors also increased uncertainty; the assumptions made within the model were another source of uncertainty; coupled with technological limitations of existing equipment**, factors like these contributed to the uncertainty. To address these uncertainties, we recommend, on one hand, that the submersible periodically send data to the mother ship, including **location information, environmental data, performance status, collected data, mission logs, and fault alarms**; on the other hand, we suggest equipping the submersible with **navigation devices, communication equipment, environmental detection sensors, sonar systems, imaging devices, and system status detection equipment** to assist in sending the aforementioned information.

Secondly, through online research, we identified a series of devices that the mother ship could carry, dividing them into communication and positioning devices, observation and detection devices, and rescue and survival devices. We utilized the

entropy-weighted TOPSIS method and grey relational analysis to rank the devices according to their scores. Selecting devices in sequence is a great choice, but given the varying functionalities, directions, and fields of expertise of the devices, we recommend selecting one device from each category. After a comprehensive evaluation, our recommended devices include **acoustic positioning systems, Autonomous Underwater Vehicles (AUV), and fully enclosed diving respirators.** Using the same method, we determined the equipment to be carried by rescue vessels: **diving support equipment, emergency rescue tools and equipment, and navigation and control systems.**

Furthermore, to achieve the goal of searching for distressed submersibles, we based our work on the previous positioning model and introduced the Bayesian search method from search theory. We divided the Ionian Sea into several large areas, which were then finely segmented into equal-sized smaller areas. By continuously updating information set D0, we combined optimal search strategies and used numerical simulation methods to test our model. To highlight the reliability of Bayesian search, we established a non-Bayesian intervention group and a Bayesian intervention group, ultimately finding that **the probability of the target's presence in new areas added after intervention was higher, thus increasing the discovery probability and achieving a 6% improvement in probability.** Moreover, we included parameters related to ocean current distribution in the probability density function of the two-dimensional normal distribution, **increasing the tail of the probability distribution in the direction of ocean currents,** making our search model more closely aligned with reality and effective.

To maximize the benefits of our model, we optimized it for application in other maritime areas. **We collected relevant information from regions such as the Caribbean Sea** and adjusted the search model parameters for use with data from different maritime areas. Additionally, to improve search efficiency, we considered the scenario of multiple submersibles searching for the same target, **introducing a networked communication cooperation mode for the submersibles.** By dividing the sea area and having multiple submersibles search separately, they could **pass signals to each other about whether the target was successfully found, using indicator function I_c and the status matrix to update the Bayesian search probabilities for stage t .** Thanks to our design of the indicator function, which exhibits a step change, the improved probability of search success in the optimized model appears very straightforward.

If you are interested in our model, we invite you to read the full text of our model development for detailed insights.

Sincerely,
MCM Team 2415009

11 Appendix

Appendix1-Tool : Python and Matlab

Appendix2-Pseudocode for Model :

Algorithm 1 Optimal Search Strategy Algorithm

Input: Initial prior distribution $P(B_{R_k} | D_0)$, v_0 , T , n , m

Output: Position of the missing submersible (x, y)

- 1: **while** No target found **do**
- 2: Initialize $Z^*(t)$ and $R^*(t)$
- 3: **for** each possible combination of $Z(t)$ and $R(t)$ **do**
- 4: $sum \leftarrow \sum_{i=1}^{mn} (1 - e^{-\frac{Z_i(t)}{\omega}}) P(B_{R_i(t)} | D_{t-1})$
- 5: **if** $Z_i^{mn}(t) < \omega$ or $\sum_{i=1}^{mn} Z_i(t) > v_0 T$ **then**
- 6: discard this combination
- 7: **end if**
- 8: $(Z^*(t), R^*(t)) \leftarrow \max\{(Z(t), R(t)), (Z^*(t), R^*(t))\}$
- 9: **end for**
- 10: Implement search and rescue in $R^*(t)$ and wait for the searching result
- 11: **if** successfully searched **then**
- 12: return (x, y)
- 13: **end if**
- 14: **if** target has not been found after reaching the maximum dwell time T **then**
- 15: $D_{t-1} \leftarrow \{\bar{A}_{R(t)}, D_{t-1}\}$
- 16: **end if**
- 17: $P(B_{R_{i(t+1)}} | D_t) \leftarrow \frac{e^{-\frac{Z_k(t)}{\omega}} P(B_{R_k} | D_{t-1})}{\sum_{R_j \in R(t)} e^{-\frac{Z_j(t)}{\omega}} P(B_{R_j} | D_{t-1}) + \sum_{R_j \notin R(t)} P(B_{R_j} | D_{t-1})}$
- 18: $P(A_{R(t)} | D_{t-1}) \leftarrow \sum_{i=1}^{mn} (1 - e^{-\frac{Z_i(t)}{\omega}}) P(B_{R_{i(t)}} | D_{t-1})$
- 19: $T \leftarrow T + 1$
- 20: **end while**

Report on Use of AI

OpenAI *ChatGPT* (Nov 5, 2023 version, ChatGPT-4)

Query1: Honey, if you want to find data on marine geographic information, which websites should you go to?

Output: When looking for data for marine geographic information, there are several well-known resources and databases you can access:

1. **National Oceanic and Atmospheric Administration (NOAA)**: NOAA provides a wide range of ocean data, including ocean maps, meteorological data, and information on marine ecosystems.
2. **World Ocean Database (WOD)**: As an important resource for international ocean research, it provides data on sea temperature, salinity and other chemical and physical parameters.
3. **Marine GIS**: These systems usually integrate a variety of marine data, such as seabed topography, biodiversity and protected areas.
4. **Google Earth and Google Ocean**: These tools provide detailed views from a global perspective to specific ocean areas, including seafloor topography.
5. **European Ocean Observation and Data Network (EMODnet)**: Provides a series of ocean data in European waters, including seafloor topography, biological and chemical data.
6. **Global Ocean Observing System (GOOS)**: is an international cooperation network established to improve the availability of ocean data.

These suggested websites provide broad coverage from basic geographical information to advanced scientific research data. For the most accurate and up-to-date data, it is recommended to explore these websites and databases directly. If you need further help accessing specific data or have other questions, please feel free to let me know.

For more information, check out these [search results](<https://chat.openai.com/backend-api/bing/redirect?query=marine%20geographic%20information%20data%20sources>).

OpenAI *ChatGPT* (Nov 5, 2023 version, ChatGPT-4)

Query 2 : Well, I want to find data on ocean dem, ocean currents, and seawater density.

Output: To find specific data such as ocean DEMs (Digital Elevation Models), ocean currents, and seawater density, you can visit the following websites:

1. **National Oceanic and Atmospheric Administration (NOAA)**:
 - NOAA's Digital Ocean Data Center provides a variety of ocean environment data, including seawater density and ocean current information.
 - NOAA's Coastal Mapping Survey Database also contains DEM data for ocean and coastal areas.
2. **European Ocean Observation and Data Network (EMODnet)**:

- EMODnet provides seafloor topographic data, including DEM.
- At the same time, it also provides data on the physical environment of the ocean, which may include ocean currents and seawater density information.

3. ****Global Ocean Data Observing System (Argo)**:**

- Argo is a global observing system that provides seawater temperature and salinity data collected from buoys, which can be used to calculate seawater density.

4. ****World Ocean Database (WOD)**:**

- Provides data on physical and chemical parameters such as seawater temperature and salinity, which can help study ocean currents and seawater density.

5. ****Geospatial Data Cloud**:**

- Provides a variety of geospatial data, including ocean DEMs. These data are suitable for terrain analysis and environmental studies.

6. ****Copernicus Marine Services**:**

- Provides comprehensive ocean data, including ocean physical and chemical parameters, ocean current data can also be obtained here.

When you access these resources, please pay attention to the license and format of the data to make sure they meet your research or project needs. If you have more specific data needs or other questions, please let me know and I will try to help you.

OpenAI *ChatGPT* (Nov 5, 2023 version, ChatGPT-4)

Query 3 : Can QGIS download ocean data?

Output: Yes, QGIS can be used to download and process ocean data. QGIS is an open source geographic information system (GIS) that supports a variety of data formats and services, including ocean data. You can access and download ocean data in QGIS via:

1. ****Use Online Data Sources**:** QGIS allows users to directly access and load data through Web Map Service (WMS), Web Feature Service (WFS) and other online GIS services. Many organizations and projects provide WMS/WFS services regarding the ocean, such as NOAA, EMODnet, and Copernicus Ocean Services.
2. ****Plug-ins**:** The QGIS community has developed many plug-ins that can help users access specific data sets directly from within the GIS software. For example, the "OpenLayers Plugin" allows users to load layers from Google Maps, OpenStreetMap, and other online map sources, which may contain useful oceanographic information.
3. ****Download data files**:** If you know the URL of a specific data set, you can directly download the data file (such as Shapefile, GeoTIFF, etc.) and then open it in QGIS. This applies to ocean DEMs, ocean currents, and seawater density data obtained from NOAA, EMODnet, Argo projects, etc.

4. ****Use QGIS' processing tools****: Once the data is loaded into QGIS, you can use its rich analysis and visualization tools to process the ocean data. QGIS supports multiple functions such as spatial analysis, raster analysis, and data management.

To use marine data effectively in QGIS, it is recommended that you become familiar with GIS concepts and QGIS operations, including how to load data layers, perform spatial queries, and create maps. In addition, it is important to understand the specific access protocols and data formats provided by the data source, which will help you work more efficiently in QGIS.

OpenAI *ChatGPT* (Nov 5, 2023 version, ChatGPT-4)

Query 4 : What is the range of ocean current speeds in the Ionian Sea?

Output: Current speeds in the Ionian Sea vary between locations and seasons, but are usually between 0.5 and 1.5 knots. This range can vary depending on time and location.

OpenAI *ChatGPT* (Nov 5, 2023 version, ChatGPT-4)

Query 5 : What are the coordinates of the center point of the Ionian Sea?

Output: The coordinates of the center point of the Ionian Sea are approximately 37 degrees north latitude and 20 degrees east longitude. Please note that this coordinate is a rough estimate, as the center point of the ocean may vary slightly under different definitions and coordinate systems.

OpenAI *ChatGPT* (Nov 5, 2023 version, ChatGPT-4)

Query 6 : Six degrees of freedom kinematics and physics equations

Output: Six degrees of freedom (6DoF) kinematics and dynamics involve all possible movements of an object in three-dimensional space. This includes three translational degrees of freedom (movement along the X, Y, and Z axes) and three rotational degrees of freedom (rotation about the X, Y, and Z axes), corresponding to:

- ****Translational degrees of freedom****:
- Movement along the X-axis (often called roll)
- Movement along the Y axis (often called pitch)
- Movement along the Z-axis (often called yaw)

- ****Rotational degrees of freedom****:
- Rotation around the X-axis (roll)
- Rotation around the Y axis (pitch)
- Rotation around the Z axis (yaw)

Kinematic equations

Kinematics is the study of the description of the motion of objects without regard to the forces causing the motion. In the 6DoF model, the kinematic equations describe how the position and attitude of an object change over time, but do not address the causes of these changes.

Kinematic equations typically involve position, velocity, acceleration, and the attitude angles (yaw, pitch, and roll) of an object.

Kinetic equations

Dynamics studies how forces and moments affect the motion of objects. The 6DoF dynamic equations consider the forces and moments acting on an object and how these forces and moments affect the object's translational and rotational motion. These equations are usually based on Newton's second law ($F=ma$), force equals mass times acceleration) and the basic principles of rotational dynamics ($\tau=I\alpha$), torque equals rotational inertia times angular acceleration). Construct.

Specific equation

For a specific 6DoF model, the dynamic equations can be specifically expressed as a set of differential equations, describing the motion state of the object in space. This includes:

- **Translational Motion Equation**: Describes the acceleration of the center of mass of an object along the X, Y, and Z axes, based on the total external force acting on the object and the mass of the object.
- **Rotational Equation of Motion**: Describes the angular acceleration of an object about its center of mass, based on the total external moment acting on the object and the object's moment of inertia.

These equations need to consider not only external forces (such as gravity, air resistance, thrust, etc.) and external moments, but also the mass distribution (affecting the moment of inertia) and speed (affecting aerodynamic forces) of the object.

In practical applications, such as aerospace engineering, robotics, and automotive dynamic simulation, 6DoF models are used to predict and analyze complex movements of objects in space. This requires the use of numerical methods and computer simulations to solve these equations, as they often cannot find analytical solutions.

OpenAI *ChatGPT* (Nov 5, 2023 version, ChatGPT-4)

Query 7 : If the resistance is divided into inertial hydrodynamic force and viscous hydrodynamic force, how should we formulate the formula?

Output: In fluid dynamics, total drag is usually divided into inertial (also called differential pressure) hydrodynamic forces and viscous hydrodynamic forces. Inertial hydrodynamic forces are mainly due to pressure changes caused by changes in fluid velocity, while viscous hydrodynamic forces are related to the viscous properties of the fluid. Here's how to break down the equation into these two parts:

1. **Inertial hydrodynamic force (pressure difference resistance)**:

$$D_i = \frac{1}{2} \rho C_{D,i} A v^2$$

where ρ is the fluid density, $C_{D,i}$ is the resistance coefficient related to the pressure difference, A is the headflow area of the submersible, v is the submersible velocity of the device relative to the fluid.

2. **Viscous hydrodynamic force (frictional resistance)**:**

$$D_v = \frac{1}{2} \rho C_{D,v} A v^2$$

where $C_{D,v}$ is the drag coefficient related to surface friction.

Then, the total resistance D can be expressed as the sum of the two:

$$D = D_i + D_v$$

or,

$$D = \frac{1}{2} \rho (C_{D,i} + C_{D,v}) A v^2$$

In practical applications, the inertial hydrodynamic coefficients and viscous hydrodynamic coefficients may need to be determined experimentally, empirically, or through computational fluid dynamics (CFD) simulations. These coefficients generally depend on the Reynolds number, surface roughness, and the shape and attitude of the submersible.

OpenAI *ChatGPT* (Nov 5, 2023 version, ChatGPT-4)

Query 8 : If these are the recommended order of equipment selected based on the evaluation method, and we need to select at least one from each of the three categories, while at the same time satisfying the recommended order of equipment and keeping the cost as low as possible, what method do you recommend me to use? Well, TOPSIS method: 8, 1, 9, 3, 6, 2, 5, 4, 7
Gray correlation method: 8, 1, 9, 2, 3, 6, 5, 4, 7

Output: In this case, you need to select at least one device from each of the three categories while satisfying the device recommendation order and cost constraints as much as possible. To balance these goals, several approaches can be taken:

1. Multi-objective optimization method

Multi-objective optimization methods can consider multiple objectives (such as equipment recommendation sequence and cost) simultaneously. You can use techniques such as linear programming, integer programming, or genetic algorithms to solve this problem. You need to define an optimization model that includes equipment selection as decision variables, equipment recommendation order and cost as optimization objectives.

2. Comprehensive evaluation method

You can combine the results of the TOPSIS method and the gray correlation method to obtain a comprehensive equipment recommendation order through a weighted average. Then, devices are selected manually or through a simple algorithm based on the comprehensive recommendation sequence and cost constraints.

3. Hierarchical decision-making method

First, make hierarchical selections based on the broad categories of equipment. Within each major category, simple decision rules (such as prioritizing equipment with high ranking and

low cost) are used to make selections based on the results of the TOPSIS method and gray correlation method and cost information.

Implementation suggestions

In this case, I recommend using a **multi-objective optimization approach**, especially if the relationship between cost data and equipment is complex. This approach can provide a systematic solution that ensures optimal equipment selection while satisfying all constraints.

If you are unfamiliar with multi-objective optimization or do not have the appropriate tools and resources, you may consider using the **Comprehensive Evaluation Method** or the **Hierarchical Decision Method** as simpler alternatives.

Implementation steps (take multi-objective optimization as an example):

1. **Define decision variables**: Indicate whether each device is selected.
2. **Construct objective function**: It may include maximizing the comprehensive recommendation order score and minimizing the total cost.
3. **Add constraints**: For example, select at least one device from each category, the total cost does not exceed the budget, etc.
4. **Solving the optimization model**: Solve the model using appropriate optimization software or libraries.
5. **Analysis results**: Select equipment based on the optimization results and conduct further analysis or adjustments.

For actual programming implementation, if you need specific help, such as code examples for building optimization models, you can further explain your needs and tools to use.

OpenAI *ChatGPT* (Nov 5, 2023 version, ChatGPT-4)

Query 9 : What is the search mode during the search and rescue of submersibles?

Output: During the search and rescue of the submersible, a series of standard search patterns are used, which are designed to maximize search area coverage and search efficiency.

Common search patterns include:

1. **Creeping Line Search**:
 - The searcher follows parallel straight paths, maintaining fixed intervals after each turn.
 - Suitable for larger search areas, when the last position of the submersible has been approximately determined.
2. **Sector Search**:
 - The searcher searches along a fan-shaped area centered on the last known position.
 - Suitable for situations where the starting point is accurately positioned.
3. **Grid Search**:
 - Conduct a systematic search in a grid pattern within the designated search area.
 - Suitable for when there is a more accurate initial search area.

4. **Expanding Square Search**:

- Starting from the last known position, gradually expand the search area in a spiral fashion.
- Used when the last known position is relatively accurate but the submersible may have drifted somewhat.

5. **Random Search**:

- When the last position cannot be accurately determined or the search area is too large, multiple points are randomly selected for search.
- Less efficient, but may be the only feasible approach in some cases.

6. **Trackline Search**:

- The searcher searches along possible submersible paths or currents.
- Used when inferring the likely path of a submersible.

The choice of search mode often depends on the specific circumstances of the submersible's disappearance, the ocean environment, the availability of search resources, and time factors. Search plans often combine multiple modes, adapted depending on how the search proceeds. In actual operations, search patterns can be optimized through computer simulation and professional software to improve search efficiency and the possibility of success.

OpenAI *ChatGPT* (Nov 5, 2023 version, ChatGPT-4)

Query1 0 : What is Bayesian search in mathematical modeling?

Output: Bayesian search is a method in mathematical modeling that uses Bayes' theorem to improve the efficiency of search and rescue operations. This method is particularly suitable for searches in large areas or with incomplete information, such as maritime search and rescue, missing person searches, etc. Bayesian search continuously updates the probability distribution of the search area by combining prior knowledge and continuously collected data, that is, where the probability of finding the target is highest.

Bayes' Theorem

Bayes' theorem is a formula in probability theory that is used to update or calculate the probability of an event based on new evidence related to that event. The formula is as follows:

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

in,

- $P(A|B)$ is the probability of event A occurring (posterior probability) under the condition that event B occurs.
- $P(B|A)$ is the probability of event B occurring under the condition that event A occurs.
- $P(A)$ is the prior probability of event A occurring.
- $P(B)$ is the probability of event B occurring.

Application of Bayesian Search

In search and rescue operations, Bayesian search uses this principle to continuously update the probability that a target may be located in a specific area. This process includes the following steps:

1. ****Prior probability distribution****: Based on existing knowledge (such as the last known location of the missing person, behavioral patterns, terrain conditions, etc.), set the preliminary probability that the target will appear at various locations in the search area.
2. ****Collect evidence****: Gather data through search campaigns and other sources of information (e.g., eyewitness reports, surveillance footage).
3. ****Update probability distribution****: Use the new information collected to update the probability of the target appearing in each location according to Bayes' theorem.
4. ****Guiding search decisions****: Based on the updated probability distribution, determine the priority areas for the next search, and give priority to those areas with a higher probability of target occurrence.

This approach makes the search process more systematic and efficient, as it ensures that search resources are focused on areas where the target is most likely to be found. As the search process progresses and new data is added, the search area is continuously adjusted to gradually narrow down the possible locations of the target.